

This blueprint details the specifications, system architectures, and flows for the subsequent developmental milestones of the Wanderwise travel assistant platform.

Step 2: Smart Time Scheduling & Transit Integration

1. Core Features & Business Logic

- * **Duration-Based Slots:** Transitioning from generic tags (Morning/Afternoon/Evening) to precise timestamps (e.g. 09:00 AM - 10:30 AM) calculated based on real monument stay statistics.
- * **Real-World Constraints:** Integrating local database/Google search queries to automatically verify monument opening/closing hours and suggest optimal weather-appropriate visit times.
- * **Automatic Transit Cards:** Pre-scheduling hotel departure transits before 09:00 AM, route travel cards in-between attractions, and return transits after the final activity.
- * **Live Delay Re-routing:** Chatbot delay detection (e.g. 'late by 45 mins') recalculates timeline shifts, checks for schedule clashes, and presents visual detour recommendations.

2. Frontend Layout & User Flow

- * **Interactive Timeline:** Timeline rendering showing cab/walk transit icons, route duration tags, and warning symbols next to delayed schedule items.
- * **Comparison Modal Overlay:** Shows side-by-side snapshots of the Original Schedule vs. the Proposed Shifted Schedule with an Approve/Reject control bar.

3. Database & API Changes

- * **Activity Schema:** Adding startTime (String), endTime (String), durationMinutes (Number), and openingHours (Object) fields.
- * **Delay Endpoint:** Creating POST /api/v1/trips/:id/delay endpoint returning a pending ChangeSuggestion document.

Step 3: Budgeting, Expense Tracking & Analytics

1. Core Features & Business Logic

- * **Planning wizard (Page 4):** Collects target budget range, traveler counts, and detailed exclusions (e.g. 'prefer street food, no fine dining') at trip creation.
- * **AI Budget Watchdog:** Detour recommendations automatically compute the cost of replacement tickets/transit and alert if the remaining budget is breached.

* **Post-Trip Financial Report:** Swaps active dashboards into structured expense reports once the trip is marked Completed.

2. Frontend Layout & User Flow

* **Log Expense Panel:** A lightweight drawer allowing users to quickly input expense amount, category, and date.

3. Database & API Changes

* **Trip & Expense Schemas:** Adding budget fields to Trip schema, and creating an Expense collection with category and description fields.

* **Analytics Endpoint:** GET /api/v1/trips/:id/expenses/analytics returning categorized sums and daily spend curves.